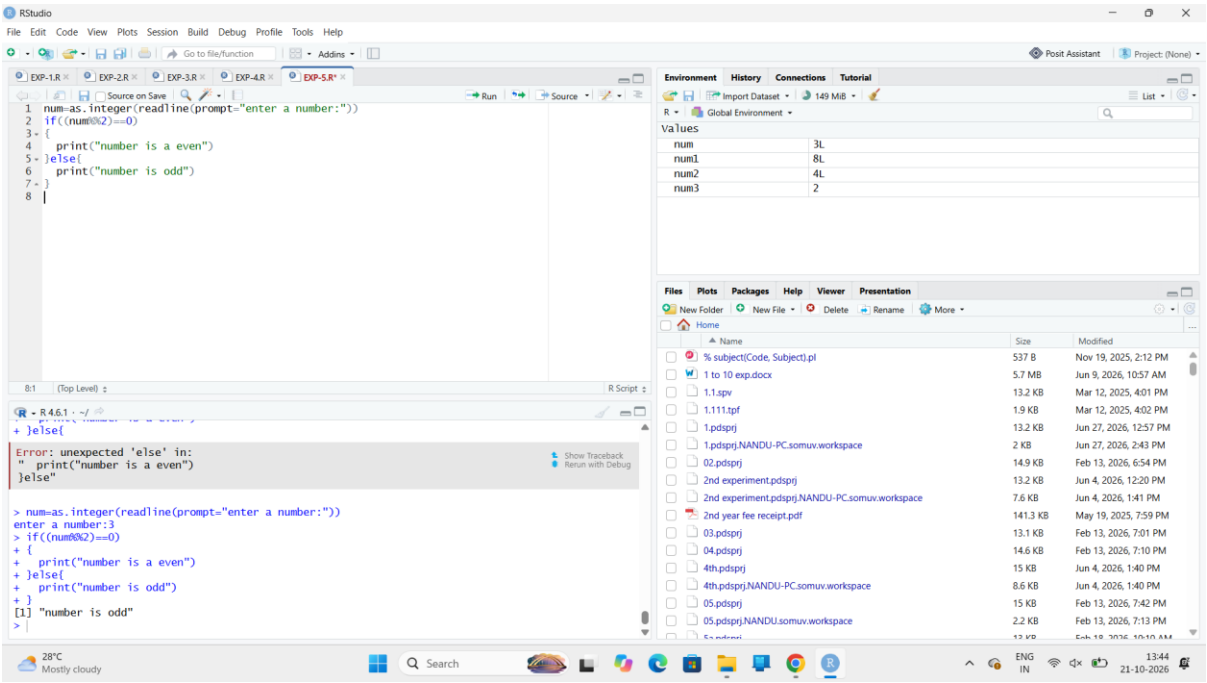
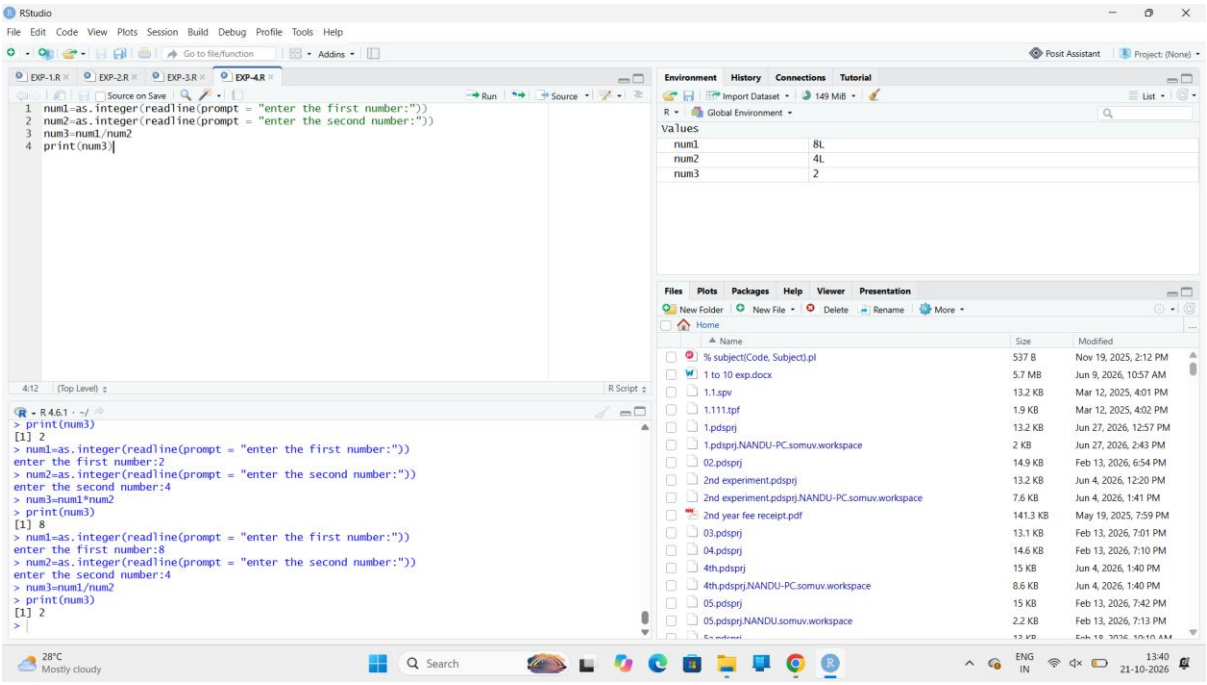
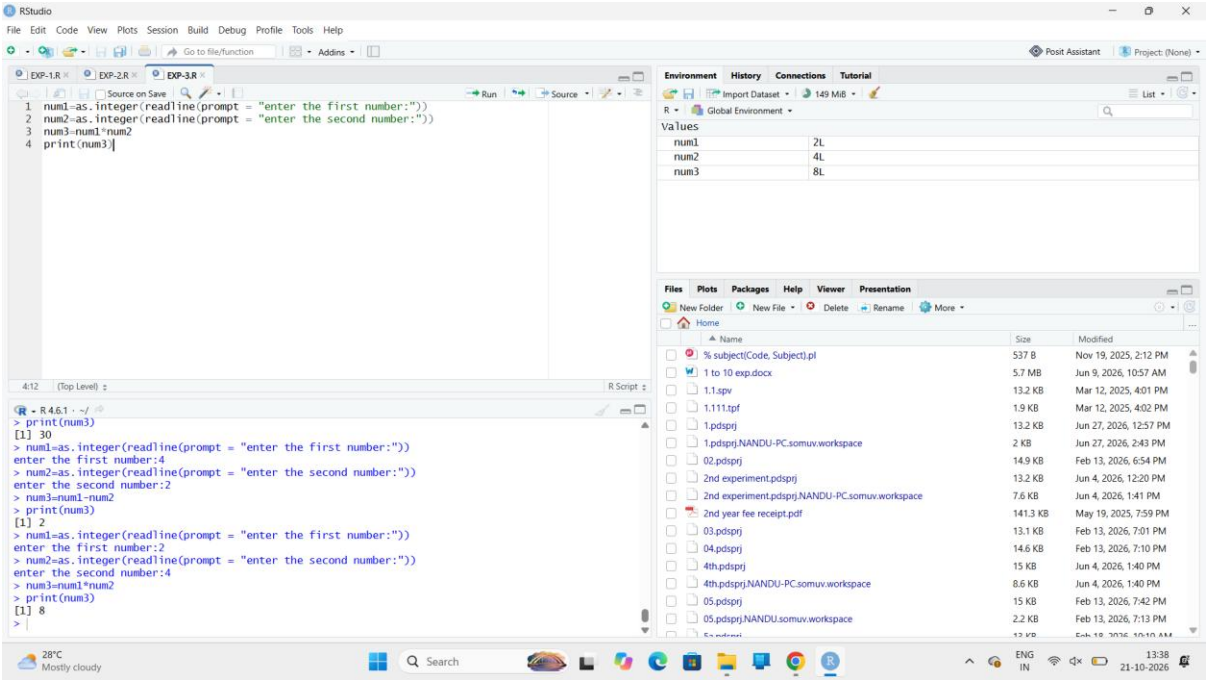
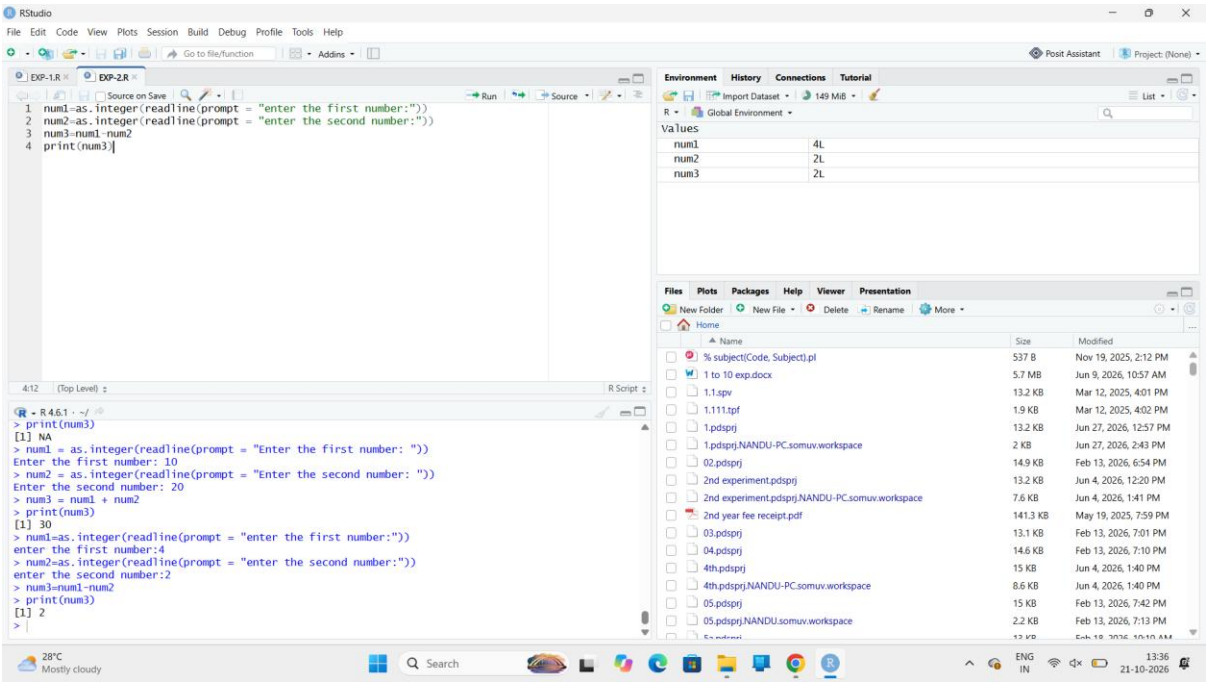
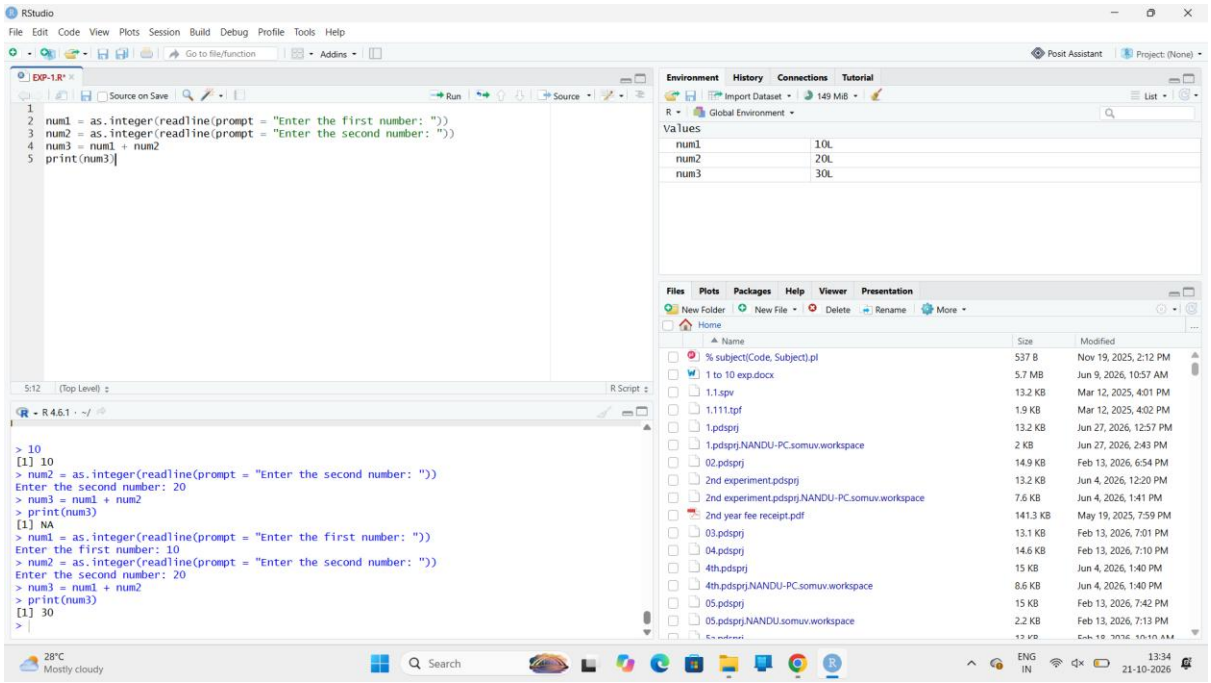
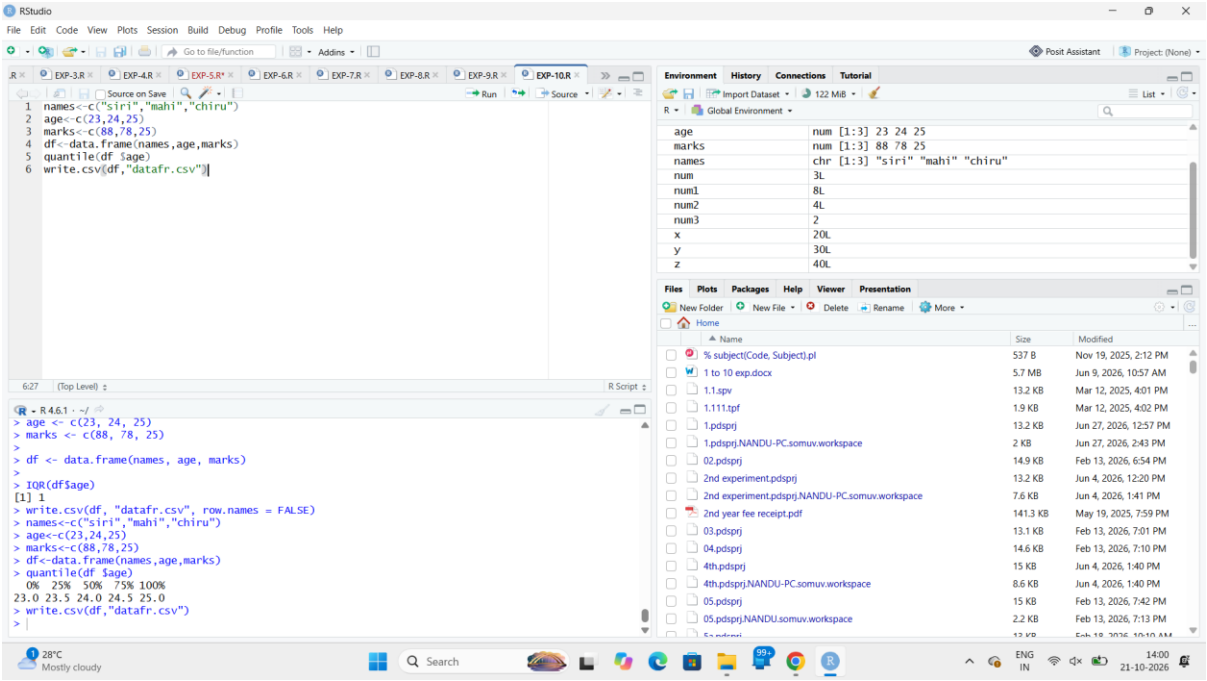

RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
EXP-1.R EXP-2.R EXP-3.R EXP-4.R EXP-5.R*
num=as.integer(readline(prompt="enter a number:"))
if((num%%2)==0)
{
print("number is a even")
}else{
print("number is odd")
}
Environment History Connections Tutorial
Values
num 3L
num1 8L
num2 4L
num3 2
Files Plots Packages Help Viewer Presentation
Error: unexpected 'else' in:
" print("number is a even")
}else"
> num=as.integer(readline(prompt="enter a number:"))
enter a number:3
> if((num%%2)==0)
+ {
+ print("number is a even")
+ }else{
+ print("number is odd")
+ }
[1] "number is odd"
28°C
Mostly cloudy
Search
ENG
IN
13:44
21-10-2026

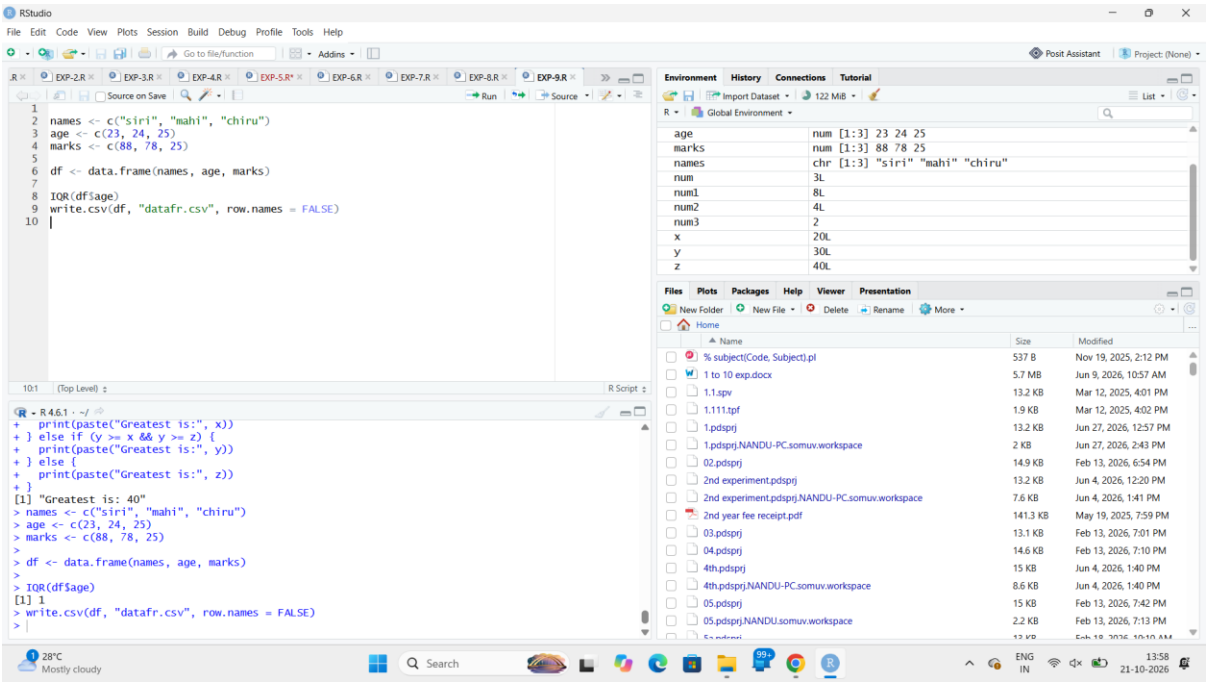

RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
EXP-1.R EXP-2.R EXP-3.R EXP-4.R
num1=as.integer(readline(prompt = "enter the first number:"))
num2=as.integer(readline(prompt = "enter the second number:"))
num3=num1/num2
print(num3)
Environment History Connections Tutorial
Values
num1 8L
num2 4L
num3 2
Files Plots Packages Help Viewer Presentation
> print(num3)
[1] 2
> num1=as.integer(readline(prompt = "enter the first number:"))
enter the first number:2
> num2=as.integer(readline(prompt = "enter the second number:"))
enter the second number:4
> num3=num1*num2
> print(num3)
[1] 8
> num1=as.integer(readline(prompt = "enter the first number:"))
enter the first number:8
> num2=as.integer(readline(prompt = "enter the second number:"))
enter the second number:4
> num3=num1/num2
> print(num3)
[1] 2
28°C
Mostly cloudy
Search
ENG
IN
13:40
21-10-2026

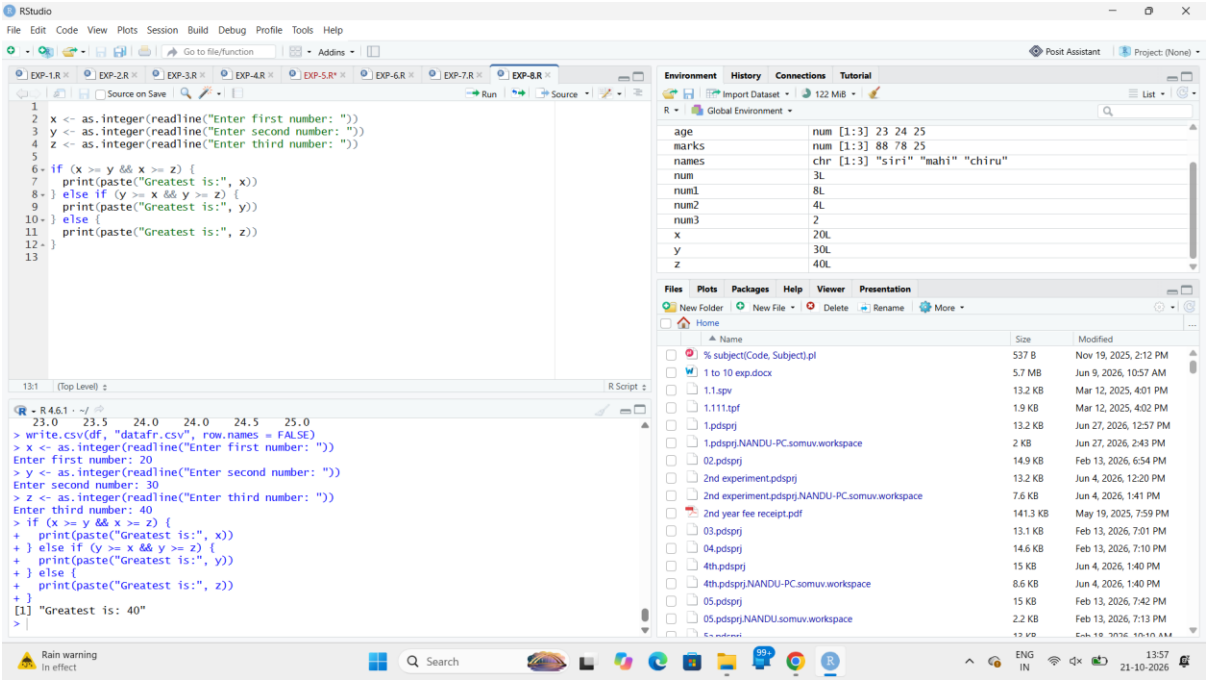

RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
EXP-1.R
EXP-2.R
EXP-3.R
num1=as.integer(readline(prompt = "enter the first number:"))
num2=as.integer(readline(prompt = "enter the second number:"))
num3=num1*num2
print(num3)
Environment History Connections Tutorial
Values
num1 2L
num2 4L
num3 8L
Files Plots Packages Help Viewer Presentation
> print(num3)
[1] 30
> num1=as.integer(readline(prompt = "enter the first number:"))
enter the first number:4
> num2=as.integer(readline(prompt = "enter the second number:"))
enter the second number:2
> num3=num1-num2
> print(num3)
[1] 2
> num1=as.integer(readline(prompt = "enter the first number:"))
enter the first number:2
> num2=as.integer(readline(prompt = "enter the second number:"))
enter the second number:4
> num3=num1*num2
> print(num3)
[1] 8
28°C
Mostly cloudy
Search
ENG
IN
13:38
21-10-2026

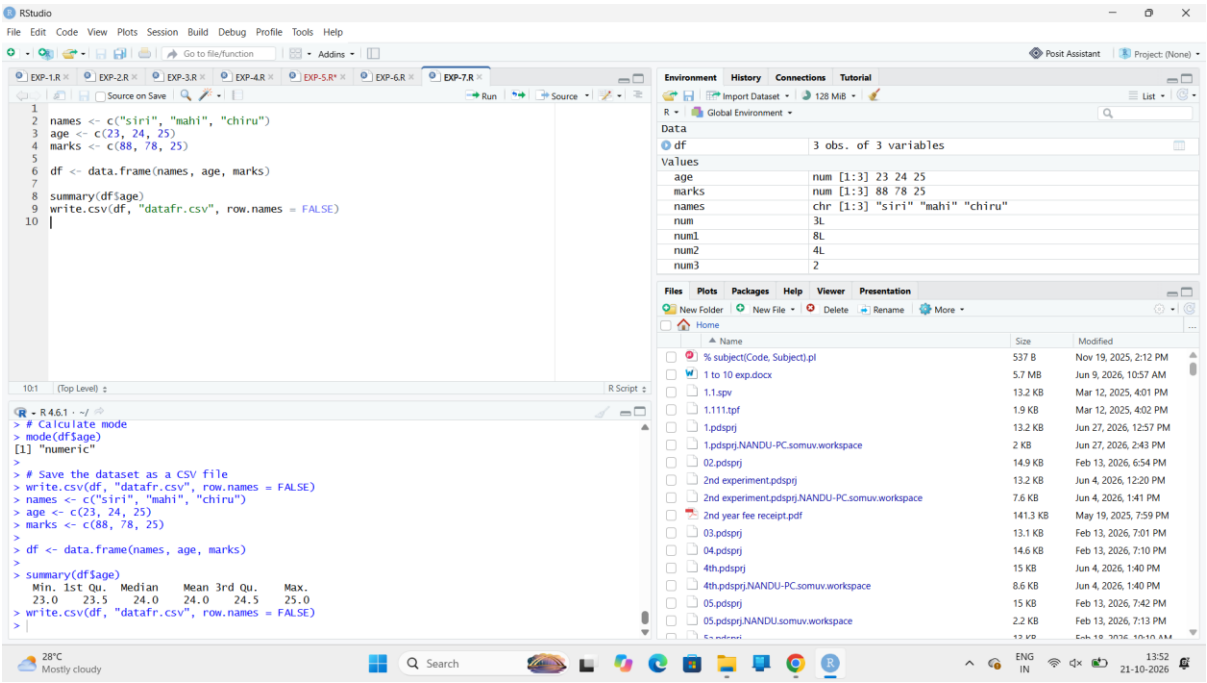

RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
EXP-1.R
EXP-2.R
num1=as.integer(readline(prompt = "enter the first number:"))
num2=as.integer(readline(prompt = "enter the second number:"))
num3=num1-num2
print(num3)
Environment History Connections Tutorial
Values
num1 4L
num2 2L
num3 2L
Files Plots Packages Help Viewer Presentation
> print(num3)
[1] NA
> num1 = as.integer(readline(prompt = "Enter the first number: "))
Enter the first number: 10
> num2 = as.integer(readline(prompt = "Enter the second number: "))
Enter the second number: 20
> num3 = num1 + num2
> print(num3)
[1] 30
> num1=as.integer(readline(prompt = "enter the first number:"))
enter the first number:4
> num2=as.integer(readline(prompt = "enter the second number:"))
enter the second number:2
> num3=num1-num2
> print(num3)
[1] 2
28°C
Mostly cloudy
Search
ENG
IN
13:36
21-10-2026


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
EXP-1.R*
1
2 num1 = as.integer(readline(prompt = "Enter the first number: "))
3 num2 = as.integer(readline(prompt = "Enter the second number: "))
4 num3 = num1 + num2
5 print(num3)
5:12 (Top Level) R Script
R 4.6.1 · ~/
> 10
[1] 10
> num2 = as.integer(readline(prompt = "Enter the second number: "))
Enter the second number: 20
> num3 = num1 + num2
> print(num3)
[1] NA
> num1 = as.integer(readline(prompt = "Enter the first number: "))
Enter the first number: 10
> num2 = as.integer(readline(prompt = "Enter the second number: "))
Enter the second number: 20
> num3 = num1 + num2
> print(num3)
[1] 30
>
Environment History Connections Tutorial
Import Dataset 149 MiB
R Global Environment
Values
num1 10L
num2 20L
num3 30L
Files Plots Packages Help Viewer Presentation
New Folder New File Delete Rename More
Home
Name Size Modified
% subject(Code, Subject).pl 537 B Nov 19, 2025, 2:12 PM
1 to 10 exp.docx 5.7 MB Jun 9, 2026, 10:57 AM
1.1.spv 13.2 KB Mar 12, 2025, 4:01 PM
1.111.tpf 1.9 KB Mar 12, 2025, 4:02 PM
1.pdsprj 13.2 KB Jun 27, 2026, 12:57 PM
1.pdsprj.NANDU-PC.somuv.workspace 2 KB Jun 27, 2026, 2:43 PM
02.pdsprj 14.9 KB Feb 13, 2026, 6:54 PM
2nd experiment.pdsprj 13.2 KB Jun 4, 2026, 12:20 PM
2nd experiment.pdsprj.NANDU-PC.somuv.workspace 7.6 KB Jun 4, 2026, 1:41 PM
2nd year fee receipt.pdf 141.3 KB May 19, 2025, 7:59 PM
03.pdsprj 13.1 KB Feb 13, 2026, 7:01 PM
04.pdsprj 14.6 KB Feb 13, 2026, 7:10 PM
4th.pdsprj 15 KB Jun 4, 2026, 1:40 PM
4th.pdsprj.NANDU-PC.somuv.workspace 8.6 KB Jun 4, 2026, 1:40 PM
05.pdsprj 15 KB Feb 13, 2026, 7:42 PM
05.pdsprj.NANDU.somuv.workspace 2.2 KB Feb 13, 2026, 7:13 PM
28°C
Mostly cloudy
Search
ENG
IN
13:34
21-10-2026


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
EXP-10.R
1 names<-c("siri","mahi","chiru")
2 age<-c(23,24,25)
3 marks<-c(88,78,25)
4 df<-data.frame(names,age,marks)
5 quantile(df $age)
6 write.csv(df,"datafr.csv")
Environment History Connections Tutorial
age num [1:3] 23 24 25
marks num [1:3] 88 78 25
names chr [1:3] "siri" "mahi" "chiru"
num 3L
num1 8L
num2 4L
num3 2
x 20L
y 30L
z 40L
Files Plots Packages Help Viewer Presentation
% subject(Code, Subject).pl 537 B Nov 19, 2025, 2:12 PM
1 to 10 exp.docx 5.7 MB Jun 9, 2026, 10:57 AM
1.1.spv 13.2 KB Mar 12, 2025, 4:01 PM
1.111.tpf 1.9 KB Mar 12, 2025, 4:02 PM
1.pdsprj 13.2 KB Jun 27, 2026, 12:57 PM
1.pdsprj.NANDU-PC.somuv.workspace 2 KB Jun 27, 2026, 2:43 PM
02.pdsprj 14.9 KB Feb 13, 2026, 6:54 PM
2nd experiment.pdsprj 13.2 KB Jun 4, 2026, 12:20 PM
2nd experiment.pdsprj.NANDU-PC.somuv.workspace 7.6 KB Jun 4, 2026, 1:41 PM
2nd year fee receipt.pdf 141.3 KB May 19, 2025, 7:59 PM
03.pdsprj 13.1 KB Feb 13, 2026, 7:01 PM
04.pdsprj 14.6 KB Feb 13, 2026, 7:10 PM
4th.pdsprj 15 KB Jun 4, 2026, 1:40 PM
4th.pdsprj.NANDU-PC.somuv.workspace 8.6 KB Jun 4, 2026, 1:40 PM
05.pdsprj 15 KB Feb 13, 2026, 7:42 PM
05.pdsprj.NANDU.somuv.workspace 2.2 KB Feb 13, 2026, 7:13 PM
R 4.6.1 · ~/
> age <- c(23, 24, 25)
> marks <- c(88, 78, 25)
> df <- data.frame(names, age, marks)
> IQR(df$age)
[1] 1
> write.csv(df, "datafr.csv", row.names = FALSE)
> names<-c("siri","mahi","chiru")
> age<-c(23,24,25)
> marks<-c(88,78,25)
> df<-data.frame(names,age,marks)
> quantile(df $age)
0% 25% 50% 75% 100%
23.0 23.5 24.0 24.5 25.0
> write.csv(df,"datafr.csv")
28°C Mostly cloudy
Search
ENG IN 14:00 21-10-2026


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
EXP-9.R
1 names <- c("siri", "mahi", "chiru")
2 age <- c(23, 24, 25)
3 marks <- c(88, 78, 25)
4 df <- data.frame(names, age, marks)
5 IQR(df$age)
6 write.csv(df, "datafr.csv", row.names = FALSE)
Environment History Connections Tutorial
age num [1:3] 23 24 25
marks num [1:3] 88 78 25
names chr [1:3] "siri" "mahi" "chiru"
num 3L
num1 8L
num2 4L
num3 2
x 20L
y 30L
z 40L
Files Plots Packages Help Viewer Presentation
% subject(Code, Subject).pl 537 B Nov 19, 2025, 2:12 PM
1 to 10 exp.docx 5.7 MB Jun 9, 2026, 10:57 AM
1.1.spv 13.2 KB Mar 12, 2025, 4:01 PM
1.111.tpf 1.9 KB Mar 12, 2025, 4:02 PM
1.pdsprj 13.2 KB Jun 27, 2026, 12:57 PM
1.pdsprj.NANDU-PC.somuv.workspace 2 KB Jun 27, 2026, 2:43 PM
02.pdsprj 14.9 KB Feb 13, 2026, 6:54 PM
2nd experiment.pdsprj 13.2 KB Jun 4, 2026, 12:20 PM
2nd experiment.pdsprj.NANDU-PC.somuv.workspace 7.6 KB Jun 4, 2026, 1:41 PM
2nd year fee receipt.pdf 141.3 KB May 19, 2025, 7:59 PM
03.pdsprj 13.1 KB Feb 13, 2026, 7:01 PM
04.pdsprj 14.6 KB Feb 13, 2026, 7:10 PM
4th.pdsprj 15 KB Jun 4, 2026, 1:40 PM
4th.pdsprj.NANDU-PC.somuv.workspace 8.6 KB Jun 4, 2026, 1:40 PM
05.pdsprj 15 KB Feb 13, 2026, 7:42 PM
05.pdsprj.NANDU.somuv.workspace 2.2 KB Feb 13, 2026, 7:13 PM
R 4.6.1 · ~/
+ print(paste("Greatest is:", x))
+ } else if (y >= x && y >= z) {
+ print(paste("Greatest is:", y))
+ } else {
+ print(paste("Greatest is:", z))
+ }
[1] "Greatest is: 40"
> names <- c("siri", "mahi", "chiru")
> age <- c(23, 24, 25)
> marks <- c(88, 78, 25)
> df <- data.frame(names, age, marks)
> IQR(df$age)
[1] 1
> write.csv(df, "datafr.csv", row.names = FALSE)
28°C Mostly cloudy
Search
ENG IN 13:58 21-10-2026


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
EXP-1.R EXP-2.R EXP-3.R EXP-4.R EXP-5.R* EXP-6.R EXP-7.R EXP-8.R
x <- as.integer(readline("Enter first number: "))
y <- as.integer(readline("Enter second number: "))
z <- as.integer(readline("Enter third number: "))
if (x >= y && x >= z) {
print(paste("Greatest is:", x))
} else if (y >= x && y >= z) {
print(paste("Greatest is:", y))
} else {
print(paste("Greatest is:", z))
}
Environment History Connections Tutorial
age num [1:3] 23 24 25
marks num [1:3] 88 78 25
names chr [1:3] "siri" "mahi" "chiru"
num 3L
num1 8L
num2 4L
num3 2
x 20L
y 30L
z 40L
Files Plots Packages Help Viewer Presentation
23.0 23.5 24.0 24.0 24.5 25.0
> write.csv(df, "datafr.csv", row.names = FALSE)
> x <- as.integer(readline("Enter first number: "))
Enter first number: 20
> y <- as.integer(readline("Enter second number: "))
Enter second number: 30
> z <- as.integer(readline("Enter third number: "))
Enter third number: 40
> if (x >= y && x >= z) {
+ print(paste("Greatest is:", x))
+ } else if (y >= x && y >= z) {
+ print(paste("Greatest is:", y))
+ } else {
+ print(paste("Greatest is:", z))
+ }
[1] "Greatest is: 40"
Rain warning
In effect
Search
ENG
IN
13:57
21-10-2026


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
EXP-1.R EXP-2.R EXP-3.R EXP-4.R EXP-5.R* EXP-6.R EXP-7.R
names <- c("siri", "mahi", "chiru")
age <- c(23, 24, 25)
marks <- c(88, 78, 25)
df <- data.frame(names, age, marks)
summary(df$age)
write.csv(df, "datafr.csv", row.names = FALSE)
Environment History Connections Tutorial
Data
df 3 obs. of 3 variables
Values
age num [1:3] 23 24 25
marks num [1:3] 88 78 25
names chr [1:3] "siri" "mahi" "chiru"
num 3L
num1 8L
num2 4L
num3 2
Files Plots Packages Help Viewer Presentation
> # Calculate mode
> mode(df$age)
[1] "numeric"
> # Save the dataset as a CSV file
> write.csv(df, "datafr.csv", row.names = FALSE)
> names <- c("siri", "mahi", "chiru")
> age <- c(23, 24, 25)
> marks <- c(88, 78, 25)
> df <- data.frame(names, age, marks)
> summary(df$age)
Min. 1st Qu. Median Mean 3rd Qu. Max.
23.0 23.5 24.0 24.0 24.5 25.0
> write.csv(df, "datafr.csv", row.names = FALSE)
28°C
Mostly cloudy
Search
ENG
IN
13:52
21-10-2026

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

EXP-1.R | EXP-2.R | EXP-3.R | EXP-4.R | EXP-5.R* | EXP-6.R | EXP-7.R

```
1
2  # Create the dataset
3  names <- c("siri", "mahi", "chiru")
4  age <- c(23, 24, 25)
5  marks <- c(88, 78, 25)
6
7  # Create a data frame
8  df <- data.frame(names, age, marks)
9
10 # Calculate mean
11 mean(df$age)
12
13 # Calculate median
14 median(df$age)
15
16 # Calculate mode
17 mode(df$age)
18
19 # Save the dataset as a CSV file
20 write.csv(df, "datafr.csv", row.names = FALSE)
21
```

8:36 (Top Level) R Script

Environment | History | Connections | Tutorial

Import Dataset | 128 MiB | Global Environment

Data

df	3 obs. of 3 variables

Values

age	num [1:3] 23 24 25
marks	num [1:3] 88 78 25
names	chr [1:3] "siri" "mahi" "chiru"
num	3L
num1	8L
num2	4L
num3	2

Files | Plots | Packages | Help | Viewer | Presentation

New Folder | New File | Delete | Rename | More

Home

Name	Size	Modified
% subject(Code, Subject).pl	537 B	Nov 19, 2025, 2:12 PM
1 to 10 exp.docx	5.7 MB	Jun 9, 2026, 10:57 AM
1.1.spv	13.2 KB	Mar 12, 2025, 4:01 PM
1.111.tpf	1.9 KB	Mar 12, 2025, 4:02 PM
1.pdsprj	13.2 KB	Jun 27, 2026, 12:57 PM
1.pdsprj.NANDU-PC.somuv.workspace	2 KB	Jun 27, 2026, 2:43 PM
02.pdsprj	14.9 KB	Feb 13, 2026, 6:54 PM
2nd experiment.pdsprj	13.2 KB	Jun 4, 2026, 12:20 PM
2nd experiment.pdsprj.NANDU-PC.somuv.workspace	7.6 KB	Jun 4, 2026, 1:41 PM
2nd year fee receipt.pdf	141.3 KB	May 19, 2025, 7:59 PM
03.pdsprj	13.1 KB	Feb 13, 2026, 7:01 PM
04.pdsprj	14.6 KB	Feb 13, 2026, 7:10 PM
4th.pdsprj	15 KB	Jun 4, 2026, 1:40 PM
4th.pdsprj.NANDU-PC.somuv.workspace	8.6 KB	Jun 4, 2026, 1:40 PM
05.pdsprj	15 KB	Feb 13, 2026, 7:42 PM
05.pdsprj.NANDU.somuv.workspace	2.2 KB	Feb 13, 2026, 7:13 PM

R 4.4.1 · ~/

```
> age <- c(23, 24, 25)
> marks <- c(88, 78, 25)
>
> # Create a data frame
> df <- data.frame(names, age, marks)
>
> # Calculate mean
> mean(df$age)
[1] 24
>
> # Calculate median
> median(df$age)
[1] 24
>
> # Calculate mode
> mode(df$age)
[1] "numeric"
>
```

28°C Mostly cloudy | Search | ENG IN | 13:54 21-10-2026

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

EXP-7.R | EXP-8.R | EXP-9.R | EXP-10.R | EXP-11.R | EXP-12.R | EXP-13.R | EXP-14.R

```
1
2  a <- c(55, 67, 89, 80, 90)
3
4  # Vertical bar plot
5  barplot(a,
6          main = "Vertical Bar Plot",
7          xlab = "Items",
8          ylab = "Values")
9
10 # Horizontal bar plot
11 barplot(a,
12         horiz = TRUE,
13         main = "Horizontal Bar Plot",
14         xlab = "Values",
15         ylab = "Items")
16
```

16:1 (Top Level) R Script

Environment | History | Connections | Tutorial

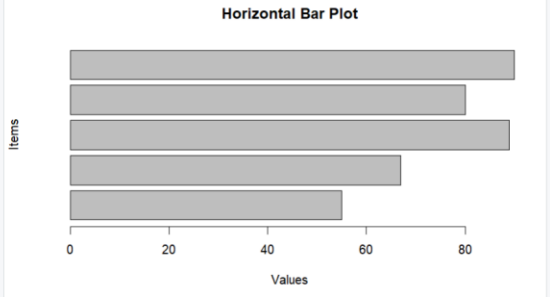
Import Dataset | 121 MiB | Global Environment

Values

a	num [1:5] 55 67 89 80 90
A	num [1:3] 23 24 25
age	num [1:3] 23 24 25
marks	num [1:3] 88 78 25
Maximum	25
Mean	24
mid_range	24
Minimum	23
MinMax	num [1:3] 0 0.5 1

Files | Plots | Packages | Help | Viewer | Presentation

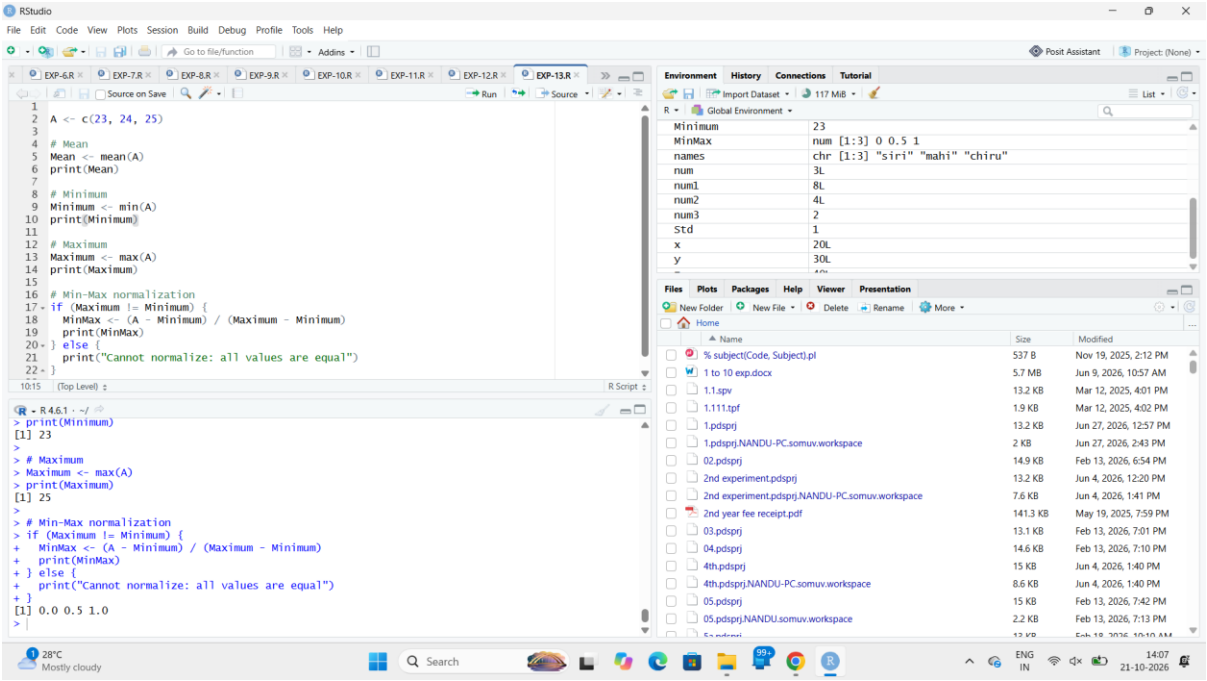
Zoom | Fit to Pane | Export | Publish

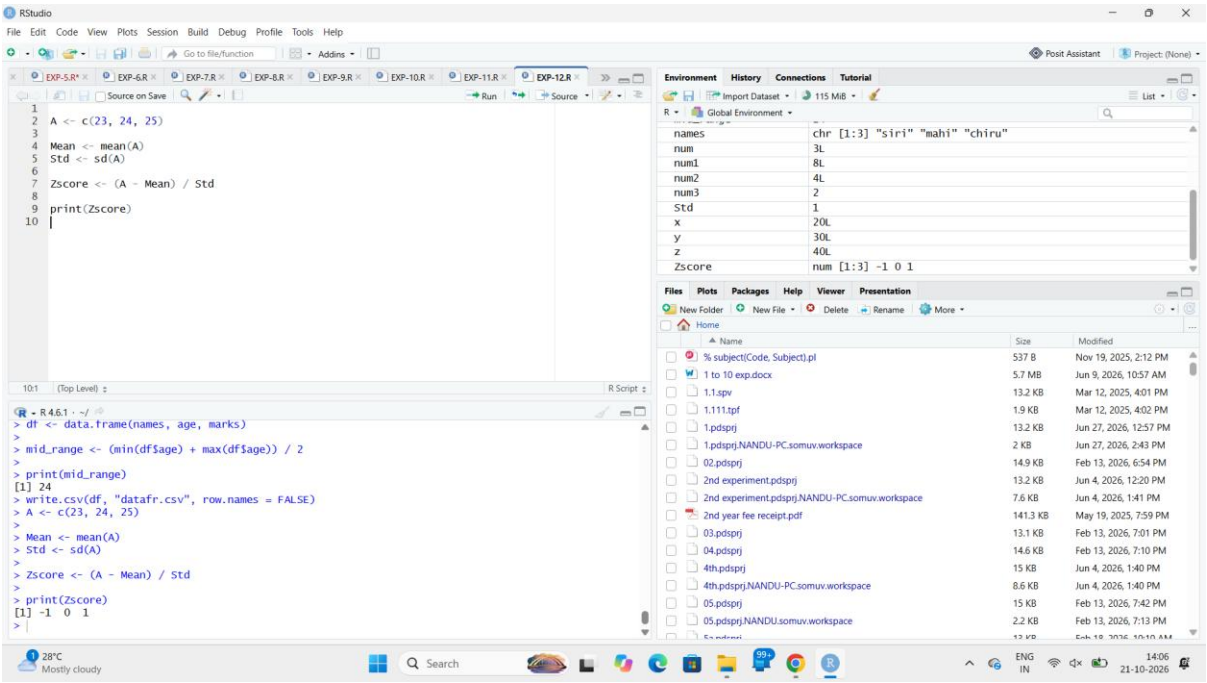


R 4.4.1 · ~/

```
+ )
[1] 0.0 0.5 1.0
> a <- c(55, 67, 89, 80, 90)
>
> # Vertical bar plot
> barplot(a,
+         main = "Vertical Bar Plot",
+         xlab = "Items",
+         ylab = "Values")
>
> # Horizontal bar plot
> barplot(a,
+         horiz = TRUE,
+         main = "Horizontal Bar Plot",
+         xlab = "Values",
+         ylab = "Items")
>
```

28°C Mostly cloudy | Search | ENG IN | 14:08 21-10-2026


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
EXP-13.R
A <- c(23, 24, 25)
Mean
Mean <- mean(A)
print(Mean)
Minimum
Minimum <- min(A)
print(Minimum)
Maximum
Maximum <- max(A)
print(Maximum)
Min-Max normalization
if (Maximum != Minimum) {
MinMax <- (A - Minimum) / (Maximum - Minimum)
print(MinMax)
} else {
print("Cannot normalize: all values are equal")
}
> print(Minimum)
[1] 23
> # Maximum
> Maximum <- max(A)
> print(Maximum)
[1] 25
> # Min-Max normalization
> if (Maximum != Minimum) {
+ MinMax <- (A - Minimum) / (Maximum - Minimum)
+ print(MinMax)
+ } else {
+ print("Cannot normalize: all values are equal")
+ }
[1] 0.0 0.5 1.0
Environment History Connections Tutorial
Minimum 23
MinMax num [1:3] 0 0.5 1
names chr [1:3] "siri" "mahi" "chiru"
num 3L
num1 8L
num2 4L
num3 2
Std 1
x 20L
y 30L
Files Plots Packages Help Viewer Presentation
% subject(Code, Subject).pl 537 B Nov 19, 2025, 2:12 PM
1 to 10 exp.docx 5.7 MB Jun 9, 2026, 10:57 AM
1.1.spv 13.2 KB Mar 12, 2025, 4:01 PM
1.111.tpf 1.9 KB Mar 12, 2025, 4:02 PM
1.pdsprj 13.2 KB Jun 27, 2026, 12:57 PM
1.pdsprj.NANDU-PC.somuv.workspace 2 KB Jun 27, 2026, 2:43 PM
02.pdsprj 14.9 KB Feb 13, 2026, 6:54 PM
2nd experiment.pdsprj 13.2 KB Jun 4, 2026, 12:20 PM
2nd experiment.pdsprj.NANDU-PC.somuv.workspace 7.6 KB Jun 4, 2026, 1:41 PM
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03.pdsprj 13.1 KB Feb 13, 2026, 7:01 PM
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4th.pdsprj 15 KB Jun 4, 2026, 1:40 PM
4th.pdsprj.NANDU-PC.somuv.workspace 8.6 KB Jun 4, 2026, 1:40 PM
05.pdsprj 15 KB Feb 13, 2026, 7:42 PM
05.pdsprj.NANDU.somuv.workspace 2.2 KB Feb 13, 2026, 7:13 PM
28°C Mostly cloudy
Search
ENG IN
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RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
EXP-12.R
A <- c(23, 24, 25)
Mean <- mean(A)
Std <- sd(A)
Zscore <- (A - Mean) / Std
print(Zscore)
> df <- data.frame(names, age, marks)
> mid_range <- (min(df$age) + max(df$age)) / 2
> print(mid_range)
[1] 24
> write.csv(df, "datafr.csv", row.names = FALSE)
> A <- c(23, 24, 25)
> Mean <- mean(A)
> Std <- sd(A)
> Zscore <- (A - Mean) / Std
> print(Zscore)
[1] -1 0 1
Environment History Connections Tutorial
names chr [1:3] "siri" "mahi" "chiru"
num 3L
num1 8L
num2 4L
num3 2
Std 1
x 20L
y 30L
z 40L
Zscore num [1:3] -1 0 1
Files Plots Packages Help Viewer Presentation
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4th.pdsprj 15 KB Jun 4, 2026, 1:40 PM
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05.pdsprj 15 KB Feb 13, 2026, 7:42 PM
05.pdsprj.NANDU.somuv.workspace 2.2 KB Feb 13, 2026, 7:13 PM
28°C Mostly cloudy
Search
ENG IN
14:06 21-10-2026

```
names <- c("siri", "mahi", "chiru")
age <- c(23, 24, 25)
marks <- c(88, 78, 25)

df <- data.frame(names, age, marks)

mid_range <- (min(df$age) + max(df$age)) / 2

print(mid_range)
write.csv(df, "datafr.csv", row.names = FALSE)
```

```
> df<-data.frame(names,age,marks)
> quantile(df $age)
  0%  25%  50%  75% 100%
23.0 23.5 24.0 24.5 25.0
> write.csv(df,"datafr.csv")
> names <- c("siri", "mahi", "chiru")
> age <- c(23, 24, 25)
> marks <- c(88, 78, 25)
>
> df <- data.frame(names, age, marks)
>
> mid_range <- (min(df$age) + max(df$age)) / 2
>
> print(mid_range)
[1] 24
> write.csv(df, "datafr.csv", row.names = FALSE)
>
```

```
# Sample dataset
diabetes <- data.frame(
  Age = c(20, 25, 30, 35, 40),
  BloodPressure = c(70, 75, 78, 85, 88)
)

# Build linear regression model
Relation <- lm(BloodPressure ~ Age, data = diabetes)

# Display model summary
print(summary(Relation))

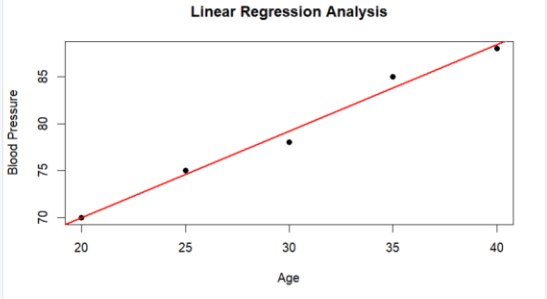
# Plot the data
plot(diabetes$Age, diabetes$BloodPressure,
     main = "Linear Regression Analysis",
     xlab = "Age",
     ylab = "Blood Pressure",
     pch = 19)

# Add regression line
abline(Relation, col = "red", lwd = 2)
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.033 on 3 degrees of freedom
Multiple R-squared:  0.9851,    Adjusted R-squared:  0.9801
F-statistic: 198.4 on 1 and 3 DF,  p-value: 0.0007752

>
> # Plot the data
> plot(diabetes$Age, diabetes$BloodPressure,
+      main = "Linear Regression Analysis",
+      xlab = "Age",
+      ylab = "Blood Pressure",
+      pch = 19)
>
> # Add regression line
> abline(Relation, col = "red", lwd = 2)
>
```



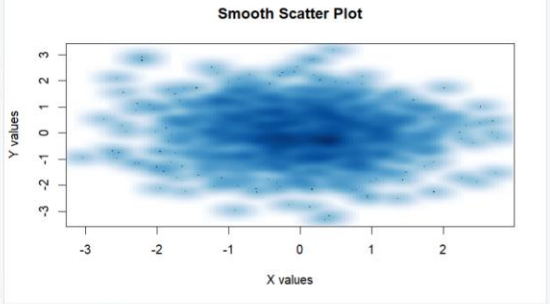
```r
set.seed(9)

x <- rnorm(1000)
y <- rnorm(1000)

# Scatter plot
plot(x, y,
     main = "Scatter Plot",
     xlab = "X values",
     ylab = "Y values",
     pch = 19)

# Smooth scatter plot, as in the PDF
smoothScatter(x, y,
              main = "Smooth Scatter Plot",
              xlab = "X values",
              ylab = "Y values")
```

```
> x <- rnorm(1000)
> y <- rnorm(1000)
>
> # Scatter plot
> plot(x, y,
+      main = "Scatter Plot",
+      xlab = "X values",
+      ylab = "Y values",
+      pch = 19)
>
> # Smooth scatter plot, as in the PDF
> smoothScatter(x, y,
+               main = "Smooth Scatter Plot",
+               xlab = "X values",
+               ylab = "Y values")
>
```




```r
AgeGroup <- c("Young", "Young", "Young",
              "Older", "Older", "Older")

InsulinGroup <- c("Low", "High", "Low",
                  "Low", "High", "High")

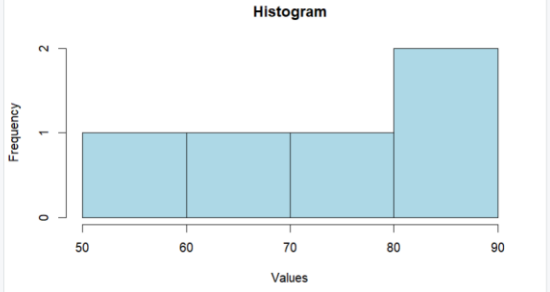
# Create frequency table
data_table <- table(AgeGroup, InsulinGroup)

print(data_table)

# Perform chi-square test
print(chisq.test(data_table))
```

```
> InsulinGroup <- c("Low", "High", "Low",
+                   "Low", "High", "High")
>
> # Create frequency table
> data_table <- table(AgeGroup, InsulinGroup)
>
> print(data_table)
        InsulinGroup
AgeGroup High Low
   Older    2   1
   Young    1   2
>
> # Perform chi-square test
> print(chisq.test(data_table))

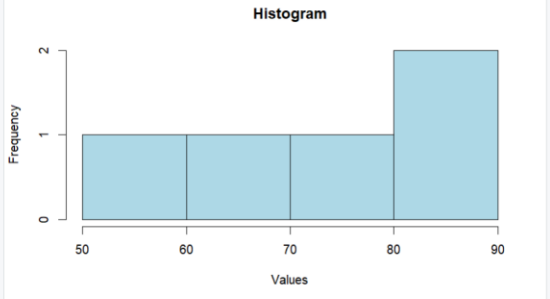
        Pearson's Chi-squared test with Yates' continuity correction
```



```
a <- c(55, 67, 89, 80, 90)

hist(a,
     main = "Histogram",
     xlab = "Values",
     ylab = "Frequency",
     col = "lightblue",
     border = "black")
```

```
> df <- data.frame(names, age, marks)
>
> # Box plot
> boxplot(df$age,
+         main = "Box Plot of Age",
+         xlab = "Age",
+         col = "lightblue")
> a <- c(55, 67, 89, 80, 90)
>
> hist(a,
+      main = "Histogram",
+      xlab = "Values",
+      ylab = "Frequency",
+      col = "lightblue",
+      border = "black")
>
```

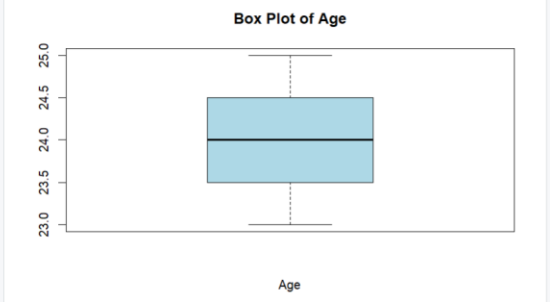



```
names <- c("siri", "chiru", "loki")
age <- c(23, 24, 25)
marks <- c(88, 78, 25)

df <- data.frame(names, age, marks)

# Box plot
boxplot(df$age,
        main = "Box Plot of Age",
        xlab = "Age",
        col = "lightblue")
```

```
> barplot(a,
+         horiz = TRUE,
+         main = "Horizontal Bar Plot",
+         xlab = "Values",
+         ylab = "Items")
> names <- c("siri", "chiru", "loki")
> age <- c(23, 24, 25)
> marks <- c(88, 78, 25)
>
> df <- data.frame(names, age, marks)
>
> # Box plot
> boxplot(df$age,
+         main = "Box Plot of Age",
+         xlab = "Age",
+         col = "lightblue")
>
```



```
  Glucose = c(90, 95, 100, 110, 115, 125)
)

# Build multiple regression model
Model <- lm(Age ~ BloodPressure + Glucose,
            data = diabetes)

# Display model
print(Model)

# Display detailed summary
print(summary(Model))

# Extract intercept and coefficients
A <- coef(Model)[1]
xBloodPressure <- coef(Model)[2]
yGlucose <- coef(Model)[3]

print(A)
print(xBloodPressure)
print(yGlucose)
```

```
> 
> # Extract intercept and coefficients
> A <- coef(Model)[1]
> xBloodPressure <- coef(Model)[2]
> yGlucose <- coef(Model)[3]
> 
> print(A)
(Intercept)
   -16.2377
> print(xBloodPressure)
BloodPressure
   -0.9098361
> print(yGlucose)
 Glucose
1.132787
> 
```

names	chr [1:3] "siri" "chiru" "loki"
num	3L
num1	8L
num2	4L
num3	2
Std	1
x	num [1:1000] -0.767 -0.816 -0.142 -0.278 0.436 ...
xBloodPressure	Named num -0.91
y	num [1:1000] 0.299 -1.127 0.582 0.855 -0.384 ...
yGlucose	Named num 1.13

